Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to CE, EEE, CSE, EIE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Define modulation and explain the need of modulation.
b) A carrier with amplitude modulated to a depth of 50% by a sinusoidal, produces side band frequencies of 5.005 MHz and 4.995 MHz. The amplitude of each side frequency is 40V. Find the frequency and amplitude of the carrier signal. [8+7]

2.a) Draw the block diagram and explain generation of AM signal.
b) A modulating signal is a multi-tone signal given by
$m(t) = A_1 \cos w_1 t + A_2 \cos w_2 t + A_3 \cos w_3 t$.
The signal m (t) modulates a carrier Ac $\cos w_c t$. Plot the signal sided spectrum and find the bandwidth of the modulating signal. Assume that $w_3 > w_2 > w_1$ and $A_3 > A_2 > A_1$. [8+7]

3.a) Discuss the effect of modulation index on the band width of FM. Explain the generation of WBFM from NBFM with neat sketch.
b) Compare the pulse modulation systems and continuous modulation systems. [10+5]

4.a) Draw the PWM waveform and explain the generation.
b) With block diagram, explain time division multiplexing system. [8+7]

5.a) Explain the principle of Combination Switching system in Telecommunication.
b) What are the differences between Space and Time Division Switching? [9+6]

6.a) Explain hardware required for LAN.
b) What are the differences between LAN, WAN, MAN and Internet? [8+7]

7.a) A satellite is in an elliptical orbit with a perigee of 1000km and an apogee of 4000 km. Using a mean earth radius 6379=8.14 km, find the period of the orbit.
b) Draw the transmitter and receiver block diagrams of an earth station and explain.[5+10]

8.a) Explain how Mobile Internet (Mobile IP) protocol works.
b) Draw domain message format, Domain header format. [7+8]

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